

Data Quality Report

Every material data-quality issue found in the raw dataset — how it was detected, how it was treated in the Golden layer, and its quantified business impact. Reproducible SQL for every check lives in `sql/05_forensics.sql`; cleaning logic in `sql/01_clean_dimensions.sql` and `sql/02_clean_events.sql`.

- **Fact**

directly observed, not open to interpretation
- **Strong Evidence**

well-supported, not 100% certain
- **Correlation**

association, no causal claim
- **Hypothesis**

plausible, unverified

01 Summary of findings

#	ISSUE	STATUS	BUSINESS IMPACT
A	Duplicate payment rows	CONFIRMED	Rs 2.59 Cr (1.93%) of reported recovery is double-counted if raw data is used as-is
B	Payment attribution undefined	CONFIRMED	Any “recovery by campaign/channel” figure is a modeling choice, not a measurement — window choice changes the attributed campaign for the majority of payments that can be attributed at all
C	Timezone field unreliable	CONFIRMED	73.5% of accounts show >1 timezone label across their own calls; unsafe for calling-time / compliance-window analysis. Immaterial to the recovery trend.
D	Vendor / disposition drift	PARTIALLY CONFIRMED	Vendor call outcomes are statistically identical across all 15 vendors (ruled out); PTP and PROMISE_TO_PAY are the same event under two labels (confirmed)
E	Agent identity unresolvable	CONFIRMED — SEVERE	Agent-dimension attributes (name, employee code, team, vendor) cannot be trusted; only agent_id is safe to use
F	Portfolio mix shift	NOT CONFIRMED	Risk segment / loan type mix of paying accounts is stable within ±3pp every month — rules out mix shift as a driver
G	Denominator manipulation	RISK DEMONSTRATED	The same July numerator produces a contact rate of 7.8%, 22.8%, or 31.1% depending purely on which denominator is chosen
H	Borrower identity unresolvable	CONFIRMED — SEVERE	77% of distinct borrowers have internally conflicting name/phone/email/city — geography and demographic cuts carry material uncertainty
I	accounts.status vs. history	CONFIRMED — SEVERE	85.7% of accounts with a status history disagree between the dimension table and the latest logged event — no single source of truth for “is this account still active”
J	Duplicate call / WhatsApp rows	CONFIRMED	1,350 duplicate call rows (−1.48%), 600 duplicate WhatsApp event rows (−1.0%) — inflates activity/volume KPIs if uncorrected
K	August 2026 is a partial month	CONFIRMED	Extract stops 8 Aug 2026; naive trending shows a 74.8% collapse vs. July — a calendar artifact, not a business event

Findings H and I were discovered independently during this analysis; they were not on the assignment's original A–G checklist.

02 Detailed findings

A Duplicate payments

CONFIRMED

DETECTION

payment_id should be a unique event key. 500 of 25,500 rows (2.0%) share a payment_id with another row. 486 of those 500 pairs are byte-for-byte identical across every column; the other 14 differ only in payment_reference being NULL in one copy — the same underlying event either way.

A trap we found and avoided: payment_reference is not a safe de-duplication key. Grouping by it instead of payment_id would have collapsed genuinely different part-payments sharing one reference (e.g. TXN0000050468 legitimately covers three separate payments of ₹47,730.92 / ₹40,930.14 / ₹127,454.47) — discarding roughly ₹17 Cr of real recovered cash. The cleanest-looking key is not always the reliable one.

TREATMENT

De-duplicated on payment_id, keeping the copy with a non-null payment_reference. See clean_payments in sql/02_clean_events.sql.

BUSINESS IMPACT

	NAIVE (RAW)	GOLDEN (DE-DUPLICATED)
SUM(amount) where status = SUCCESS	Rs 1,341,485,926	Rs 1,315,583,965
Overstatement		Rs 25,901,962 (1.93%)

B Attribution errors — methodology risk

CONFIRMED

DETECTION

payments carries no campaign_id, call_id, or channel of its own — attributing a payment to “the campaign that caused it” requires choosing a look-back window over calls.

LOOK-BACK WINDOW	SUCCESSFUL PAYMENTS ATTRIBUTABLE TO SOME CAMPAIGN TOUCH
3 days	723 / 17,534 (4.1%)
30 days	5,535 / 17,534 (31.6%)

4,812 payments (27% of all successful payments) are attributable under a 30-day window but not a 3-day window — the attributed campaign is largely an artifact of the window an analyst picks, not a measured fact. More fundamentally, **68% of successful payments have no traceable campaign touch within 30 days at all**. Any campaign-level ROI figure built on this data is Hypothesis-strength at best.

TREATMENT

The Golden layer does not invent a payment→campaign attribution. golden_payments carries account-level context for cohort analysis but leaves campaign attribution as an explicit, parameterized step (sql/05_forensics.sql, check B) so any consumer must consciously choose and document a window.

C Timezone problems

CONFIRMED — IMMATERIAL TO TREND

DETECTION

accounts.timezone and calls.timezone each take one of three values, roughly evenly split. But 20,874 of 28,408 accounts with calls (73.5%) show calls tagged with more than one timezone label across their own history, and 60,887 of 91,350 calls (66.6%) disagree with their own account’s timezone. The field is stamped per-event, essentially at random, rather than being a stable property of location.

WHY IT DOESN’T EXPLAIN THE RECOVERY TREND

Call volume by hour-of-day is close to uniform (±3%) regardless of timezone label, so it doesn’t distort monthly recovery in this extract — but it would distort any calling-window compliance analysis (permitted hours in India are typically 8am–7pm).

TREATMENT

Carried through unchanged with a documented caveat; not used as an input to any monthly trend metric. Not recommended for compliance reporting until fixed at the source.

D Vendor mapping / disposition code drift

PARTIALLY CONFIRMED

DETECTION – VENDORS: NOT CONFIRMED

call_status is used identically by all 15 telephony vendors — ANSWERED rate is statistically indistinguishable across vendors (range 19.3%–20.5%, no outlier). schema_version (v1/v2/v3) shows no vocabulary or rate difference either. No evidence of a vendor-driven code-mapping problem.

DETECTION – DISPOSITION CODES: CONFIRMED

call_dispositions.disposition_code contains PTP and PROMISE_TO_PAY as two separate values describing the same event, co-occurring in a near-constant ~50/50 split every month (Jan: 588 vs. 566; Jul: 534 vs. 519). disposition_version does not correlate with calendar time, so it cannot signal a system migration — it looks like arbitrary per-row metadata.

TREATMENT

Canonicalized PROMISE_TO_PAY → PTP in clean_call_dispositions. Any PTP-rate metric computed on the raw filter alone understates true PTP volume by roughly half.

E Agent identity problems

CONFIRMED — SEVERE

DETECTION

agents.csv has 30,000 rows but only 1,000 distinct agent_id — the only trustworthy key, used as the foreign key by calls, call_attempts, call_dispositions, agent_sessions, field_visits, and promises_to_pay, with zero orphaned agent_ids. The ~30 rows per agent_id are not a clean change history: employee_code, agent_name, vendor_id, and team are effectively scrambled. AGT0000367 appears 48 times with 48 different employee codes and spans all 10 names in the company’s name pool. Going the other direction is no better: EMP00900 alone maps to 46 different agent_ids.

TREATMENT

agent_id is treated as the only reliable grain. clean_agents reconstructs one best-guess attribute row per agent_id using the mode of each field, tie-broken by most recent updated_at, flagged attributes_low_confidence = TRUE. Any driver analysis segmenting by agent team or vendor (from the agent dimension) should be read as directional only.

F Portfolio mix changes

NOT CONFIRMED

DETECTION

Risk segment and loan type composition of accounts making successful payments was tracked month by month. Both are flat: risk segment shares stay within LOW 24.2–27.6%, MEDIUM 22.5–26.2%, HIGH 23.1–26.8%, NPA 23.0–26.5%; loan type shares stay within 18.4–21.4% for all 5 types, every month. No evidence the business acquired a materially different portfolio at any point in the window — rules out portfolio-mix shift as a driver of any month-to-month recovery swing.

G Denominator manipulation

NOT CONFIRMED IN DATA — RISK DEMONSTRATED

DETECTION

daily_targeting — the closest table to “the population the business considers in scope” — is flat at ~6,150 accounts/month, evenly split across QUEUED/CONTACTED/SKIPPED/EXPIRED; no evidence of accounts being silently dropped over time in this table. However, we cannot see the business’s own reporting queries, only the raw event data, so downstream denominator games cannot be ruled out. To make the risk concrete, July’s contact rate was computed three defensible ways:

DENOMINATOR DEFINITION	CONTACT RATE
Answered ÷ whole portfolio (30,000)	7.8%
Answered ÷ accounts worked that month (10,278)	22.8%
Answered ÷ accounts still ACTIVE (7,539)	31.1%

The same numerator produces a 4× range purely from denominator choice. We recommend the business publish its denominator definition alongside every rate metric going forward.

H Borrower identity problems (found independently)

CONFIRMED — SEVERE

DETECTION

borrowers.csv has 30,600 rows but only 11,015 distinct borrower_id. For borrower_ids with more than one row, name/phone/email/city/state frequently disagree — up to 11 conflicting variants for a single ID, the same identity-scrambling pattern as agents (Finding E). **8,518 of 11,015 distinct borrowers (77%) have at least one internal conflict.**

WHY IT MATTERS

Geography is an explicitly required investigation dimension. With 77% of borrower records internally inconsistent on city/state, any geography-cut of recovery performance carries material uncertainty and should not be presented to leadership without this caveat.

TREATMENT

De-duplicated to one row per borrower_id (latest updated_at), with identity_conflict_flag = TRUE carried through to golden_accounts so downstream consumers can see which cuts are trustworthy.

I accounts.status vs. status history

CONFIRMED — SEVERE

DETECTION

accounts.status disagrees with the most recent status logged in account_status_history for **22,295 of 25,999 accounts with any history (85.7%)**. There is no data-level way to tell which source is authoritative — both are internally self-consistent, they simply don't agree with each other.

BUSINESS IMPACT

Any metric depending on "is this account currently active / closed / written off" — portfolio-size denominators, recovery-rate denominators, eligibility for further collections activity — is being computed against a dimension table that disagrees with the event log for the large majority of the book. **Escalated as an open question for Engineering/Product; no winner was silently picked.**

TREATMENT

Both are retained: status_snapshot (from accounts) and status_from_history (from the event log), with status_source_conflict_flag marking every disagreement.

J Duplicate call / WhatsApp event rows

CONFIRMED

DETECTION

calls: 91,350 rows / 90,000 distinct call_id (1,350 duplicate groups — 1,271 byte-for-byte exact, 79 disagree on one field, e.g. a re-logged echo).
whatsapp_events: 60,600 rows / 60,000 distinct whatsapp_event_id (600 exact duplicates).

TREATMENT

De-duplicated to one row per key (sql/02_clean_events.sql). Any activity-volume KPI computed on raw data overstates true volume by ~1.5% (calls) and ~1.0% (WhatsApp).

K August 2026 is a partial month

CONFIRMED

DETECTION

The data extract ends 8 Aug 2026. Naive month-over-month comparison shows August "declining" 74.8% vs. July — purely because August has 8 days of data instead of ~31. A calendar-completeness artifact, not a business signal.

TREATMENT

All monthly trend analysis in this project is restricted to the 7 complete months (Jan-Jul 2026); August figures are reported separately, explicitly labeled partial, never plotted on the same trend line without a visual break.

03 What we did not find

- **Portfolio mix change** (Finding F).
- **Vendor-driven code drift** (Finding D, vendor half).
- **Denominator manipulation actually occurring** in the raw targeting data (Finding G) — though the risk is real and easy to demonstrate.
- **Simpson's paradox** in the monthly recovery trend — a precondition is a shifting subgroup mix, and the subgroup mix (risk segment, loan type) is confirmed stable, so no paradox can be hiding in the aggregate.

04 Net effect on the headline recovery number

	AMOUNT
Raw, naive SUM(amount) where status = SUCCESS (whole extract)	Rs 1,341,485,926
Golden, de-duplicated SUM(amount) where status = SUCCESS (whole extract)	Rs 1,315,583,965
Total correction	Rs 25,901,962 (1.93% lower than reported)

This correction alone is *not* what explains the reported "11% improvement" — see `reports/executive_memo.docx` and `notebook/analysis.ipynb` §4 for the full explanation of that number.

Collections Analytics · Data Quality Report · Source: `collections_30k_dataset` · Full evidence and reproducible SQL: `sql/05_forensics.sql`, `sql/01_clean_dimensions.sql`, `sql/02_clean_events.sql`